

Silicon Photonic ODE Solver Simulation

Simulation and Modeling Report

Based on the Experimental Framework by Wu et al. (*Optics Express*, 2014)

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Abstract

All-optical computing architectures based on silicon-on-insulator (SOI) microring resonators (MRRs) provide an ultrafast, energy-efficient paradigm for performing mathematical operations directly on optical envelopes, eliminating electronic analog-to-digital conversion bottlenecks. Based on the seminal framework proposed by Wu et al. (*Optics Express*, 2014), this report presents a comprehensive technical evaluation of an on-chip temporal ordinary differential equation (ODE) solver equipped with thermo-optically tunable interferometric couplers. In Section 1, the fundamental governing physical equations, coupled-mode theory (CMT) formulation, and core physical quantities are established. In Section 2, the global parameter boundaries in coefficient space (a_0, b_0) achievable through microheater temperature tuning are mapped alongside the 2D operational control landscapes. In Section 3, the vulnerability of the single-cavity solver to realistic thermal noise and substrate drift ($\sigma_T = 20$ mK) is analyzed using Monte Carlo statistical simulations and precision sweeps. Finally, Section 4 examines the robustness implications of cascading two MRRs to synthesize second-order LTI differential equations, highlighting the severe performance penalties induced by inter-cavity resonance detuning and notch splitting.

1 Physics of the MRR ODE Solver and Physical Quantities

1.1 Coupled-Mode Theory and Add-Drop MRR Architecture

General linear time-invariant (LTI) physical systems driven by an arbitrary temporal input signal $x(t)$ producing an output $y(t)$ are described by constant-coefficient ordinary differential equations (ODEs). In high-bandwidth optical communications and photonic signal processing, solving these equations directly in the optical domain enables real-time computing for signals with tens of gigahertz of modulation bandwidth.

The elementary physical building block consists of an add-drop microring resonator monolithically integrated on a standard silicon-on-insulator wafer (strip waveguide cross-section 450×220 nm², group index $n_g = 4.1850$, circumference $L = 178.98$ μ m). The optical carrier is locked to a resonance wavelength $\lambda_0 = 1550.391$ nm ($\omega_0 = 2\pi c/\lambda_0 \approx 1.215 \times 10^{15}$ rad/s). The cavity round-trip transit time is $\tau_{rt} = n_g L/c \approx 2.498$ ps, corresponding to a Free Spectral Range (FSR) of $\text{FSR} = 1/\tau_{rt} \approx 400.24$ GHz ($\Delta\lambda_{\text{FSR}} \approx 3.2$ nm).

In the vicinity of resonance ω_0 , optical transmission from the input bus to the through port is accurately described via temporal Coupled-Mode Theory (CMT) as:

$$T(\omega) = \frac{j(\omega - \omega_0) + \gamma_i + \gamma_{e2} - \gamma_{e1}}{j(\omega - \omega_0) + \gamma_i + \gamma_{e1} + \gamma_{e2}} \quad (1)$$

where γ_i , γ_{e1} , and γ_{e2} denote the intrinsic cavity loss decay rate, external decay rate into the through waveguide, and external decay rate into the drop waveguide, respectively. These decay rates relate to the quality factors of the resonator according to:

$$\gamma_i = \frac{\omega_0}{2Q_i} = \frac{c \alpha_{\text{lin}}}{2n_g}, \quad \gamma_{e1} = \frac{\omega_0}{2Q_{e1}} = -\frac{c \ln(1 - \kappa_1)}{2n_g L}, \quad \gamma_{e2} = \frac{\omega_0}{2Q_{e2}} = -\frac{c \ln(1 - \kappa_2)}{2n_g L} \quad (2)$$

where $\alpha_{\text{lin}} = 100 \times \alpha_{\text{dB/cm}} \ln(10)/10 \approx 184.21$ m⁻¹ is the linear attenuation coefficient for a baseline waveguide loss $\alpha = 8.0$ dB/cm. This yields an intrinsic round-trip power loss $\eta = 1 - \exp(-\alpha_{\text{lin}} L) \approx 3.25\%$ and intrinsic quality factor $Q_i \approx 9.21 \times 10^4$. The parameters κ_1 and κ_2 represent the power coupling coefficients of the two coupling regions.

1.2 Mapping to First-Order Linear Differential Equations

Taking the inverse Fourier transform of the normalized baseband transfer function $H(\omega) = (j\omega + b_0)/(j\omega + a_0)$, the complex optical envelope of the through-port output $y(t)$ satisfies the first-order ODE:

$$\frac{dy(t)}{dt} + a_0 y(t) = \frac{dx(t)}{dt} + b_0 x(t) \quad (3)$$

Table 1: Baseline Physical, Geometric, and Operational Parameters of the SOI MRR Solver.

Parameter Symbol	Physical Meaning	Design Value	Unit
λ_0, f_0	Optical Carrier Wavelength and Frequency	1550.391 nm, 193.366	THz
L, n_g	Cavity Circumference and Group Index	178.98 μm , 4.1850	–
$\tau_{\text{rt}}, \text{FSR}$	Round-Trip Delay and Free Spectral Range	2.498 ps, 400.24	GHz
α, γ_i	Propagation Loss and Intrinsic Decay Rate	8.0 dB/cm, 6.598×10^9	rad/s
Q_i, κ_0	Intrinsic Quality Factor and Coupler Cross-Ratio	9.21×10^4 , 0.0441	–
dn/dT	Silicon Thermo-Optic Coefficient	1.86×10^{-4}	K^{-1}
L_{b1}, L_{r1}	Coupler 1 Bus and Ring Arm Physical Lengths	116.80 μm , 47.12 μm	–
L_{b2}, L_{r2}	Coupler 2 Bus and Ring Arm Physical Lengths	47.12 μm , 47.12 μm	–

where the constant coefficients a_0 and b_0 are defined as:

$$a_0 = \gamma_i + \gamma_{e1} + \gamma_{e2} = \frac{\omega_0}{2Q_{\text{total}}} = \frac{\Delta\omega_{3\text{dB}}}{2}, \quad b_0 = \gamma_i + \gamma_{e2} - \gamma_{e1} \quad (4)$$

The loaded quality factor is $Q_{\text{total}} = (Q_i^{-1} + Q_{e1}^{-1} + Q_{e2}^{-1})^{-1}$, and $\Delta\omega_{3\text{dB}}$ is the 3-dB angular bandwidth. The exact multi-pass through-port transfer function of the physical ring is:

$$H_{\text{MRR}}(\omega) = \frac{r_1 - r_2 a_{\text{loss}} e^{-j\omega\tau_{\text{rt}}}}{1 - r_1 r_2 a_{\text{loss}} e^{-j\omega\tau_{\text{rt}}}} \quad (5)$$

where $r_{1,2} = \sqrt{1 - \kappa_{1,2}}$ and $a_{\text{loss}} = \sqrt{1 - \eta} \approx 0.9836$. For pulses whose spectrum is contained within the FSR, Eq. (5) matches Eq. (1) with an error NMSE < -60 dB.

1.3 Physical Role of Coefficients and Optical Coupling Regimes

The coefficients a_0 and b_0 dictate the physical computation carried out by the device:

- **Bandwidth and Damping Coefficient (a_0):** Since $a_0 = \Delta\omega_{3\text{dB}}/2$, it governs the temporal relaxation time $\tau_{\text{cav}} = 1/a_0$ of the optical solver. For typical loaded $Q \sim 1.5 \times 10^4$, $a_0 \sim 4 \times 10^{10}$ rad/s ($\Delta f_{3\text{dB}} \approx 13$ GHz), matching picosecond optical pulse dynamics.
- **Zero Placement and Coupling Symmetry (b_0):** Governs the baseband notch depth $(b_0/a_0)^2$ and determines the physical coupling regime:
 1. *Over-coupled* ($b_0 < 0 \iff \gamma_{e1} > \gamma_i + \gamma_{e2}$): Input coupling exceeds cavity dissipation. The transmission phase undergoes a full monotonic 2π transition across the resonance.
 2. *Critically coupled* ($b_0 = 0 \iff \gamma_{e1} = \gamma_i + \gamma_{e2}$): Complete destructive cancellation occurs at carrier frequency ($T(\omega_0) = 0$). Equation (3) reduces to an *all-optical temporal differentiator*: $dy/dt + a_0 y = dx/dt$.
 3. *Under-coupled* ($b_0 > 0 \iff \gamma_{e1} < \gamma_i + \gamma_{e2}$): Losses dominate coupling; the net through-port phase across resonance returns to zero.

2 Coefficient Boundaries Achievable via Temperature Tuning

2.1 Tunable MZI Couplers and Thermo-Optic Operation

Reconfigurability of a_0 and b_0 is achieved by replacing static couplers with Mach-Zehnder Interferometric (MZI) couplers. Each MZI consists of two 3-dB-like directional couplers with power splitting ratio $\kappa_0 = 0.0441$, bounding the power coupling to $\kappa \in [0, \kappa_{\text{max}}]$, with $\kappa_{\text{max}} = 4\kappa_0(1 - \kappa_0) \approx 0.1686$. Microheaters placed above the bus arms tune the phase difference $\Delta\phi_{1,2}(T_{1,2})$ via the thermo-optic effect:

$$\kappa_{1,2}(T_{1,2}) = 2\kappa_0(1 - \kappa_0) \left[1 + \cos \left(\Delta\phi_{1,2}^{(0)} + \frac{2\pi}{\lambda_0} \frac{dn}{dT} L_{b1,2} \Delta T_{1,2} \right) \right] \quad (6)$$

Heater 1 has an asymmetric length ($L_{b1} = 116.80 \mu\text{m}$, $\Delta L_1 = 69.68 \mu\text{m}$), yielding a half-wave temperature $\Delta T_{\pi,1} \approx 35.7$ K. Heater 2 has symmetric arms ($L_{b2} = L_{r2} = 47.12 \mu\text{m}$, $\Delta L_2 = 0$), requiring $\Delta T_{\pi,2} \approx 88.5$ K.

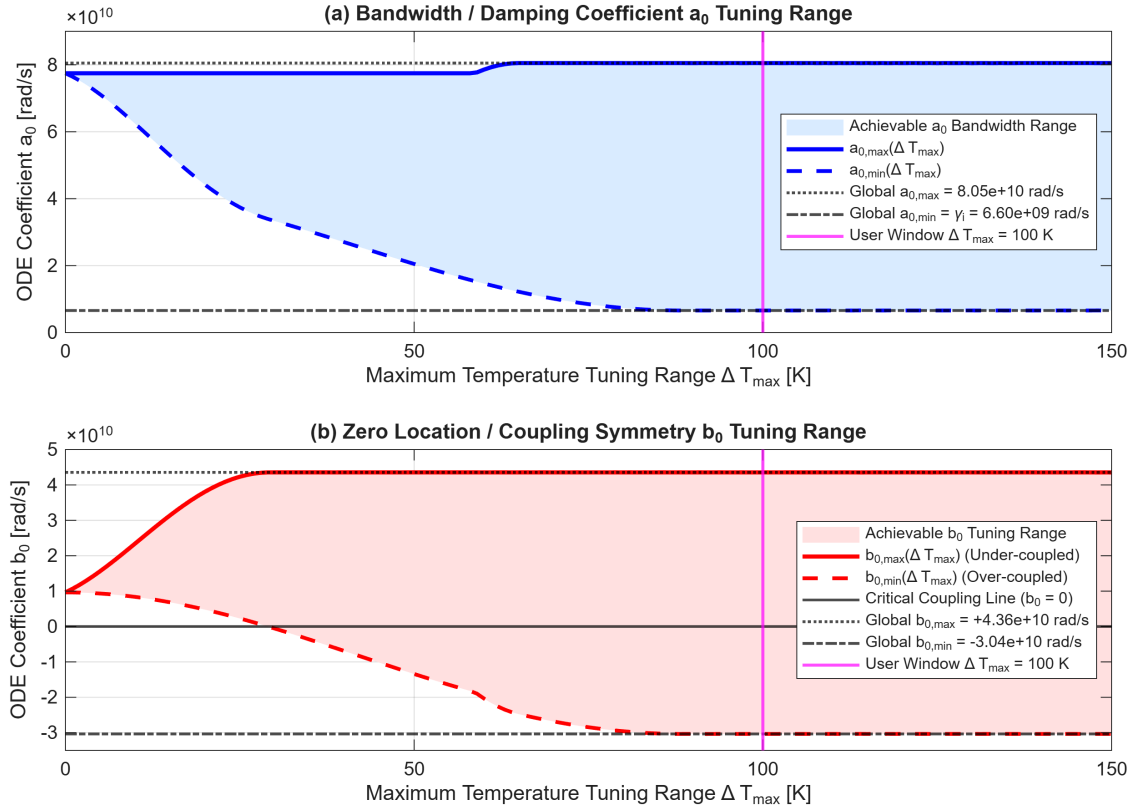


Figure 1: Continuous tuning bounds of the ODE coefficients as a function of the maximum temperature tuning elevation $\Delta T_{\max}$: (a) Bandwidth/damping coefficient a_0 , and (b) zero location coefficient b_0 , highlighting the critical coupling differentiator line ($b_0 = 0$).

2.2 Boundaries in (a_0, b_0) Space

Because $\kappa_{1,2} \in [0, \kappa_{\max}]$, the external decay rates span $\gamma_{e1,2} \in [0, \gamma_{e,\max}]$, where $\gamma_{e,\max} = -c \ln(1 - \kappa_{\max}) / (2n_g L) \approx 3.696 \times 10^{10}$ rad/s. This establishes absolute physical limits on the ODE coefficients:

$$a_{0,\min} = \gamma_i \approx 6.598 \times 10^9 \text{ rad/s}, \quad a_{0,\max} = \gamma_i + 2\gamma_{e,\max} \approx 8.052 \times 10^{10} \text{ rad/s} \quad (7)$$

$$b_{0,\min} = \gamma_i - \gamma_{e,\max} \approx -3.036 \times 10^{10} \text{ rad/s}, \quad b_{0,\max} = \gamma_i + \gamma_{e,\max} \approx +4.356 \times 10^{10} \text{ rad/s} \quad (8)$$

The continuous evolution of achievable coefficient bounds as a function of maximum temperature elevation $\Delta T_{\max}$ is shown in Fig. 1. For small thermal budgets ($\Delta T_{\max} < 30$ K), the coefficients are restricted to a narrow range. As $\Delta T_{\max}$ increases, the boundaries expand until full saturation at $\Delta T_{\max} \geq 88.5$ K, encompassing both under- and over-coupled regimes, and crossing the critical coupling differentiator region ($b_0 = 0$).

2.3 Two-Dimensional Thermal Control Landscapes

To explore simultaneous controllability of a_0 and b_0 , heater temperatures (T_1, T_2) were swept across $[20, 120]^\circ\text{C}$ ($\Delta T \leq 100$ K). Figure 2 displays the resulting 2D parameter landscapes.

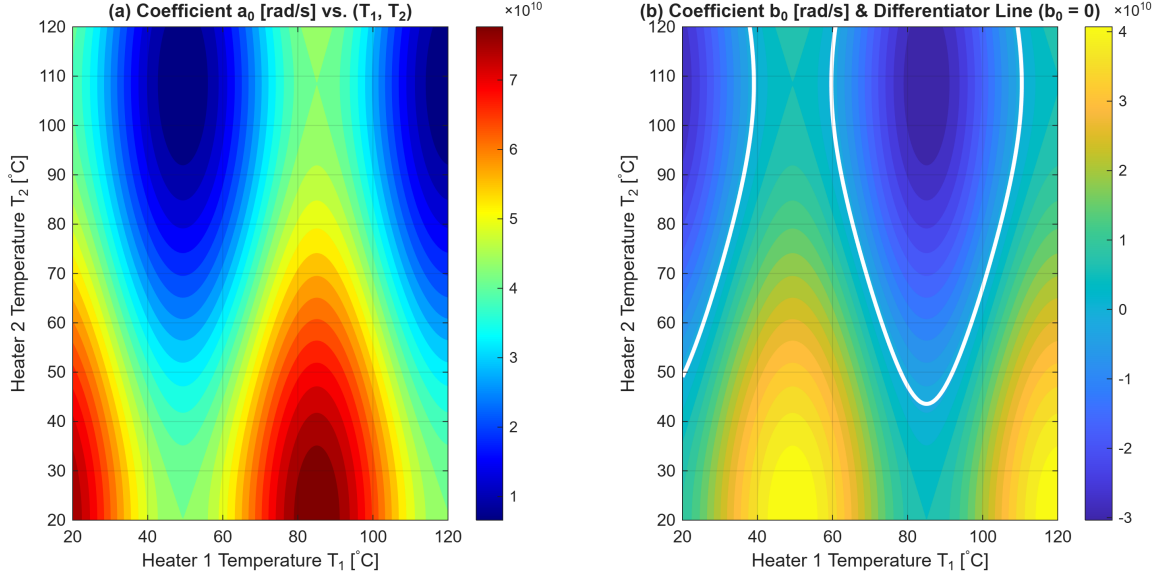


Figure 2: Two-dimensional thermal control landscapes across heater temperatures $(T_1, T_2) \in [20, 120]^\circ\text{C}$: (a) Contour map of coefficient a_0 , and (b) contour map of coefficient b_0 , highlighting the white contour of the critical differentiator condition ($b_0 = 0$).

The coefficient $a_0(T_1, T_2)$ varies smoothly over a dynamic range of $12.2\times$, from 6.6×10^9 rad/s to 8.05×10^{10} rad/s, as shown in Fig. 2(a). In Fig. 2(b), the white contour marks the critical coupling differentiator condition ($b_0 = 0$). Along this curve, $\gamma_{e1}(T_1) - \gamma_{e2}(T_2) = \gamma_i$. The existence of this continuous contour confirms that pure temporal differentiation can be synthesized across various bandwidths a_0 through coordinated multi-variable heating (T_1, T_2) .

3 Effects of Thermal Noise on Solver Performance

3.1 Physical Origins of Thermal Noise and Perturbation Modeling

Silicon's large thermo-optic coefficient ($dn/dT = 1.86 \times 10^{-4} \text{ K}^{-1}$) makes microheaters efficient, but renders the resonator sensitive to unwanted temperature fluctuations:

1. **Microheater Driver Jitter:** Multi-channel DAC quantization, current ripple, and flicker noise produce random temperature variations $\delta T_{h1}, \delta T_{h2} \sim \mathcal{N}(0, \sigma_{T, \text{heater}}^2)$ ($\sigma_{T, \text{heater}} \approx 20$ mK).
2. **Substrate and Ambient Drift:** Residual drift of the submount thermoelectric cooler (TEC) generates bulk cavity fluctuations $\delta T_{\text{sub}} \sim \mathcal{N}(0, \sigma_{T, \text{ring}}^2)$ ($\sigma_{T, \text{ring}} \approx 15$ mK).
3. **Thermal Cross-Talk:** Unguided heat diffusion from the TiN heaters leaks thermal power into the adjacent ring cavity with a coupling ratio $\xi_{\text{leak}} \approx 6\%$.

The net cavity temperature perturbation is $\delta T_{\text{cav}} = \delta T_{\text{sub}} + \xi_{\text{leak}}(\delta T_{h1} + \delta T_{h2})/2$. Variations δT_{cav} perturb the round-trip phase by $\delta \phi_{\text{rt}} = (2\pi/\lambda_0)(dn/dT)L \delta T_{\text{cav}}$, shifting the cavity resonance frequency by:

$$\Delta f_{\text{res}} = \frac{\delta \phi_{\text{rt}}}{2\pi\tau_{\text{rt}}} = \frac{dn/dT}{n_g} f_0 \delta T_{\text{cav}} \approx 8.60 \text{ GHz/K} = 8.60 \text{ MHz/mK} \quad (9)$$

For $\sigma_T = 20$ mK, Eq. (9) yields $\Delta f_{\text{res}} \approx 172$ MHz, breaking carrier symmetry and introducing dispersion.

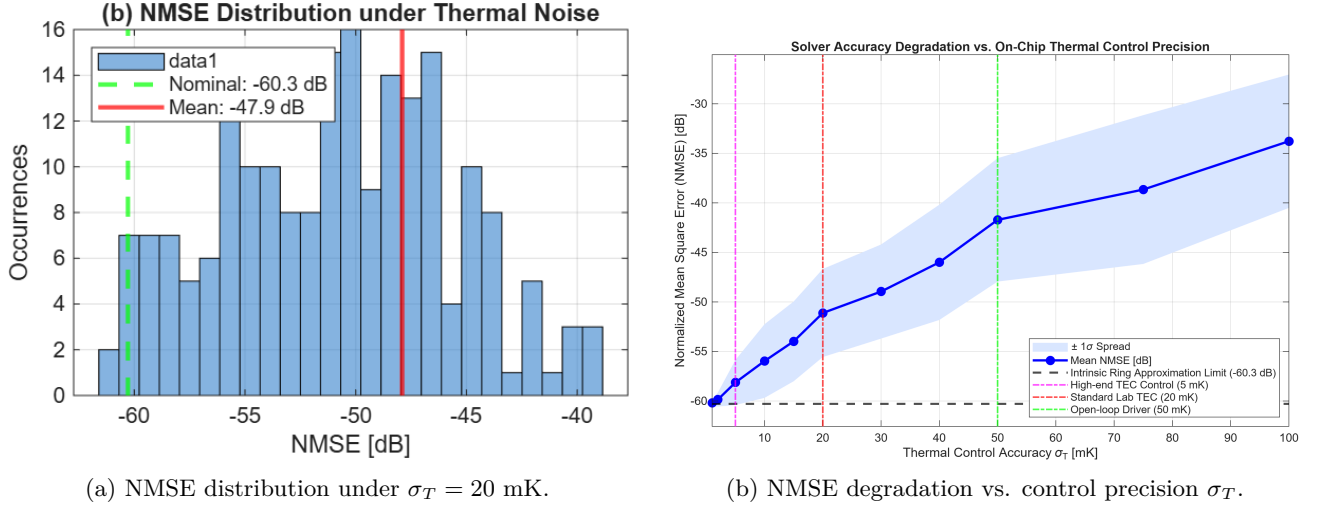


Figure 3: Single-MRR performance under thermal fluctuations: (a) 200-trial Monte Carlo NMSE histogram showing degradation from nominal (-60.3 dB) to mean (-47.9 dB); (b) Sensitivity sweep across $\sigma_T \in [1, 100]$ mK with $\pm 1\sigma$ spread and benchmark lab control regimes.

3.2 Monte Carlo Statistical Evaluation and Sensitivity Scaling

Performance degradation under thermal noise was evaluated using a 10-Gb/s Gaussian pulse (FWHM = 45 ps). Computation fidelity is quantified by Normalized Mean Square Error (NMSE) between ideal ODE output $I_{\text{ODE}}(t)$ and physical output $I_{\text{MRR}}(t)$: $\text{NMSE} = \int |I_{\text{ODE}} - I_{\text{MRR}}|^2 dt / \int |I_{\text{ODE}}|^2 dt$. As shown in Fig. 3(a), noiseless MRR accuracy reaches $\text{NMSE}_{\text{nom}} = -60.3$ dB. Under 20-mK noise, the mean error degrades by +12.4 dB to $\text{NMSE}_{\text{mean}} = -47.9$ dB. Figure 3(b) details the sensitivity across precision tiers: high-end closed-loop PID control ($\sigma_T \leq 5$ mK) preserves accuracy below -58 dB, standard lab TECs (20 mK) yield ~ -51 dB, while open-loop drivers (50 mK) induce severe distortion (> -42 dB).

4 Effects on Robustness After Cascading Two MRRs

4.1 Synthesis of Second-Order Differential Equations

Higher-order differential equations are synthesized by cascading add-drop stages in series. Connecting the through port of MRR 1 to the input bus of MRR 2 realizes a general second-order linear ODE:

$$\frac{d^2 y(t)}{dt^2} + a_1 \frac{dy(t)}{dt} + a_0 y(t) = \frac{d^2 x(t)}{dt^2} + b_1 \frac{dx(t)}{dt} + b_0 x(t) \quad (10)$$

The composite baseband transfer function is $T_{\text{ODE},2}(\omega) = T_1(\omega)T_2(\omega)$, yielding the synthesized coefficients:

$$a_1 = a_{10} + a_{20}, \quad a_0 = a_{10}a_{20}, \quad b_1 = b_{10} + b_{20}, \quad b_0 = b_{10}b_{20} \quad (11)$$

where (a_{10}, b_{10}) and (a_{20}, b_{20}) are the first-order coefficients of the individual resonators. By configuring Stage 1 in an over-coupled regime ($\kappa_{11} = 0.12, \kappa_{12} = 0.02$) and Stage 2 in an under-coupled regime ($\kappa_{21} = 0.04, \kappa_{22} = 0.06$), a non-degenerate second-order solver is formed.

4.2 Inter-Cavity Detuning and Resonance Notch Splitting

The cascaded system requires 4 independent microheaters and 2 independent resonant cavities. Uncorrelated thermal fluctuations in the two cavities produce **inter-cavity detuning**:

$$\Delta f_{\text{inter}} = |\Delta f_{\text{res},1} - \Delta f_{\text{res},2}| = \frac{dn/dT}{n_g} f_0 |\delta T_{\text{cav},1} - \delta T_{\text{cav},2}| \quad (12)$$

As illustrated in Fig. 4(a), the ideal 2nd-order response reshapes the input pulse into a primary peak accompanied by precursor and relaxation lobes. When perturbed by thermal noise ($\sigma_T = 20$ mK), instantaneous errors concentrate around the pulse transitions. In the spectral domain (Fig. 4(b)), non-zero inter-cavity detuning ($|\Delta f| = 66.0$ MHz) splits the deep quadratic second-order resonance notch into two separate dips and offsets the steep 360° phase transition across carrier frequency ω_0 .

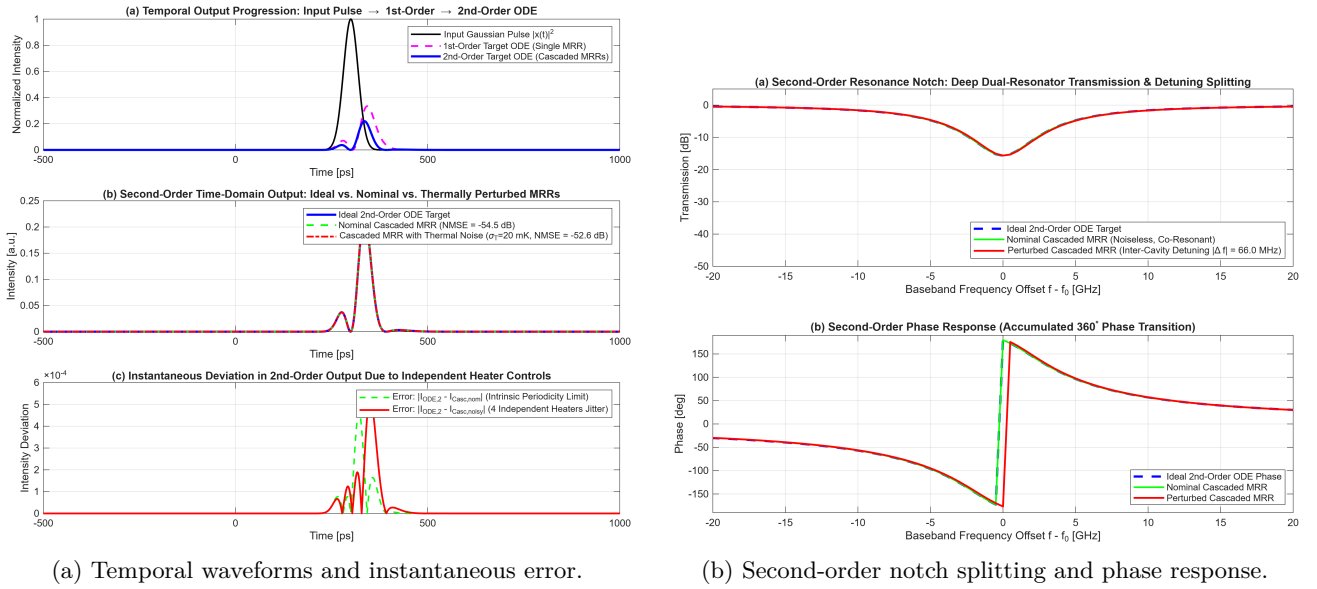


Figure 4: Cascaded 2nd-order ODE solver behavior: (a) Time-domain output progression (input $\rightarrow$ 1st-order $\rightarrow$ 2nd-order), nominal vs. thermally perturbed output, and calculation error; (b) Baseband spectral transmission and 360° phase response showing notch bifurcation under inter-cavity detuning ($|\Delta f| = 66.0$ MHz).

4.3 Statistical Robustness Comparison: Single MRR vs. Cascaded MRRs

To evaluate the stability penalty of high-order architectures, a 200-trial Monte Carlo simulation was executed under identical thermal noise conditions ($\sigma_T = 20$ mK).

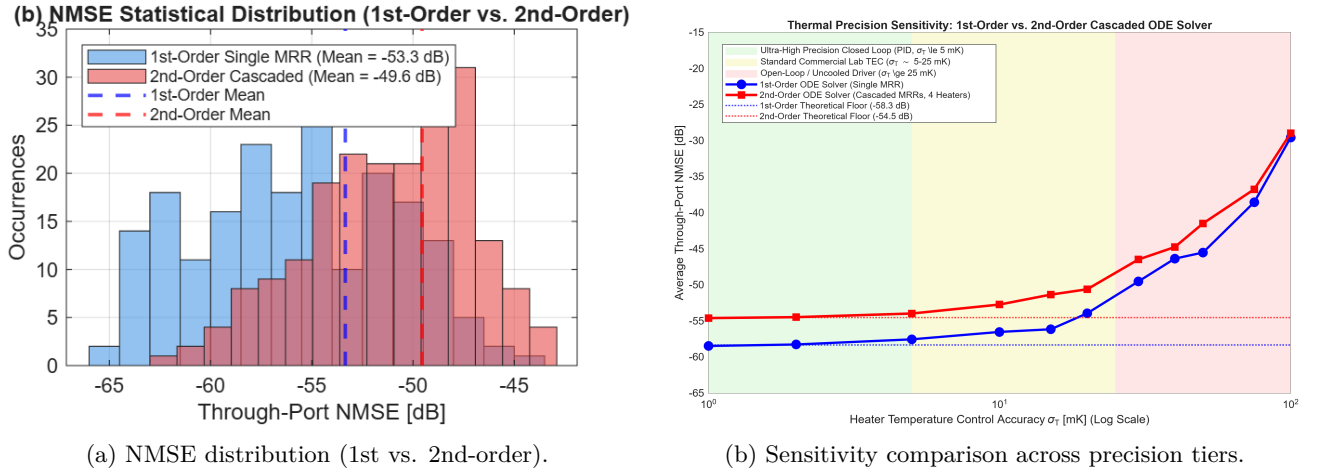


Figure 5: Statistical robustness comparison between 1st-order and 2nd-order solvers: (a) Through-port NMSE histograms under $\sigma_T = 20$ mK showing the degradation from -53.3 dB (1st-order) to -49.6 dB (2nd-order); (b) Thermal sensitivity curves across $\sigma_T \in [1, 100]$ mK showing a persistent 3 – 4 dB penalty and distribution broadening across all laboratory regimes.

As shown in Fig. 5(a), cascading two resonators incurs a $+3.7$ dB degradation in mean accuracy, shifting the average NMSE from -53.3 dB (single ring) to -49.6 dB (cascaded rings). Moreover, the distribution tail of the cascaded system broadens considerably, with worst-case trials degrading past -43 dB.

Figure 5(b) highlights the comparative sensitivity scaling across heater control precisions $\sigma_T \in [1, 100]$ mK. The theoretical noiseless approximation floor drops from -58.3 dB for the single ring to -54.5 dB for the cascaded system. Across all operational regimes (closed loop ≤ 5 mK, commercial lab TEC $5 - 25$ mK, and uncooled drivers ≥ 25 mK), the cascaded solver consistently incurs an accuracy penalty of 3 – 4 dB. This increased vulnerability stems from two compounding mechanisms:

1. **Compounding Parameter Drift:** Second-order coefficients depend multiplicatively on first-order parameters ($a_0 = a_{10}a_{20}$, $b_0 = b_{10}b_{20}$), magnifying underlying parameter variances.
2. **Resonance Notch Bifurcation:** Independent cavity drifting destroys the symmetry of the cascaded transfer function, converting the deep transmission notch into a dispersive double-trough profile.

4.4 Summary and Design Guidelines

The quantitative findings establish key engineering guidelines for on-chip photonic ODE solvers:

- **First-Order Solvers:** Highly robust against realistic thermal noise; standard commercial TEC submount control ($\sigma_T \leq 20$ mK) is sufficient to guarantee processing accuracy below $\text{NMSE} < -47$ dB.
- **Cascaded Higher-Order Solvers:** Suffer from inter-cavity detuning; maintaining NMSE below -50 dB requires high-precision closed-loop control ($\sigma_T \leq 5$ mK) or active inter-cavity resonance locking (e.g., dither-tone tracking or monolithic thermal shunts) to suppress $\Delta f_{\text{inter}} < 10$ MHz.

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